CS & IT ENGINEERING

Digital Logic

Boolean Theorems & Logical Gates

Lecture No. 07

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Recap of Previous Lecture

Topic Introduction of Digital Logic

+

$\left\{ \begin{array}{l} \text{XOR} \\ \text{XNOR} \quad \text{properties} \end{array} \right\}$

Topics to be Covered

Topic Boolean Theorem & Logic Gates

\+ Questions

Properties + Shortcuts

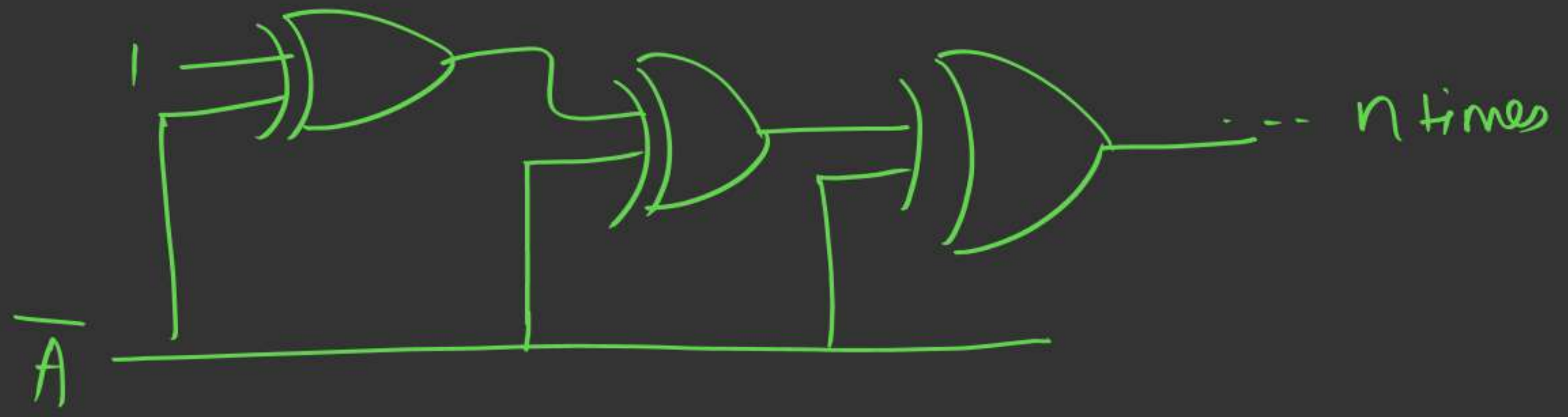
$$A \oplus (B \cdot C) = (A \oplus B)(A \oplus C) \quad \times$$

$$A \cdot (B \oplus C) = (A \cdot B) \oplus (A \cdot C) \quad \checkmark$$

$$A + (B \oplus C) = (A + B) \oplus (A + C) \quad \times$$

$$A \cdot (B \odot C) = A \cdot B \odot A \cdot C \quad \times$$

$$\left\{ \begin{aligned} A \oplus B \oplus AB &= (A+B) \\ A \oplus B \oplus (A+B) &= A \cdot B \end{aligned} \right\}$$



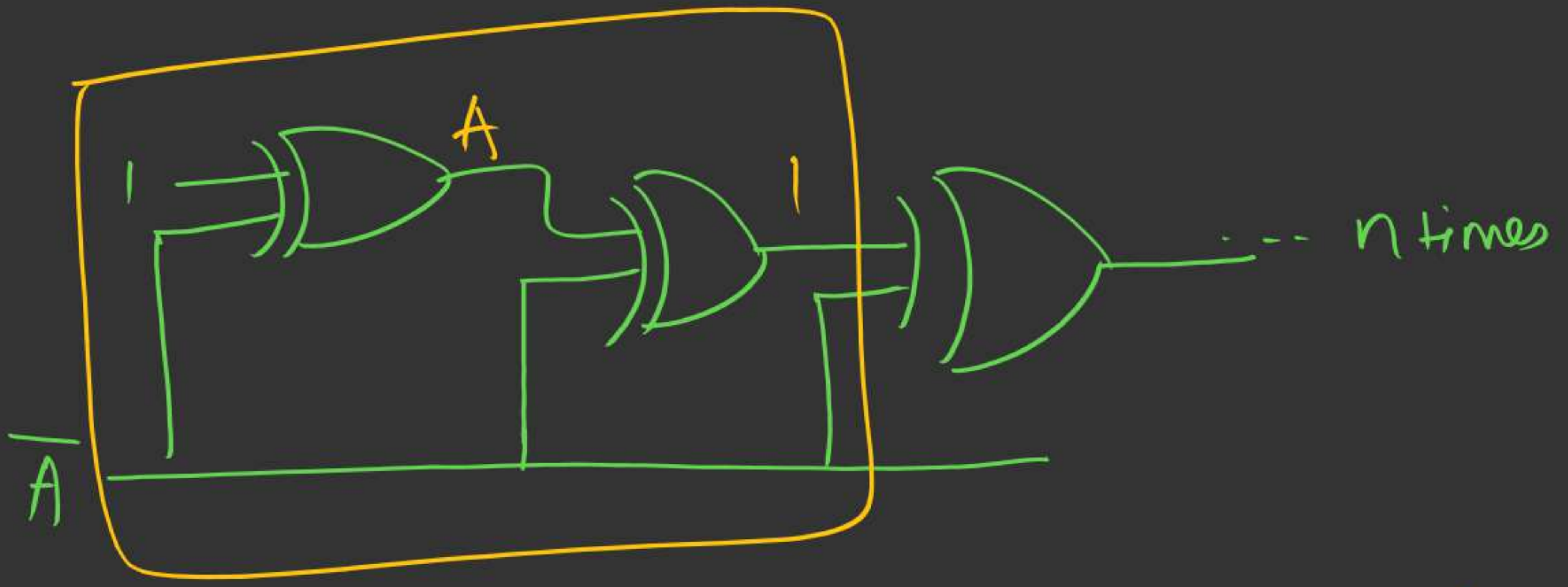
O/P :

A) A , if n = even

B) A , if n = odd

C) $\bar{A}$, if n = even

D) 1 , if n = odd.



$1 \oplus \bar{A} = \bar{\bar{A}} = A$

$A \oplus \bar{A} = 1$

o/p = 1 even

n = odd

A

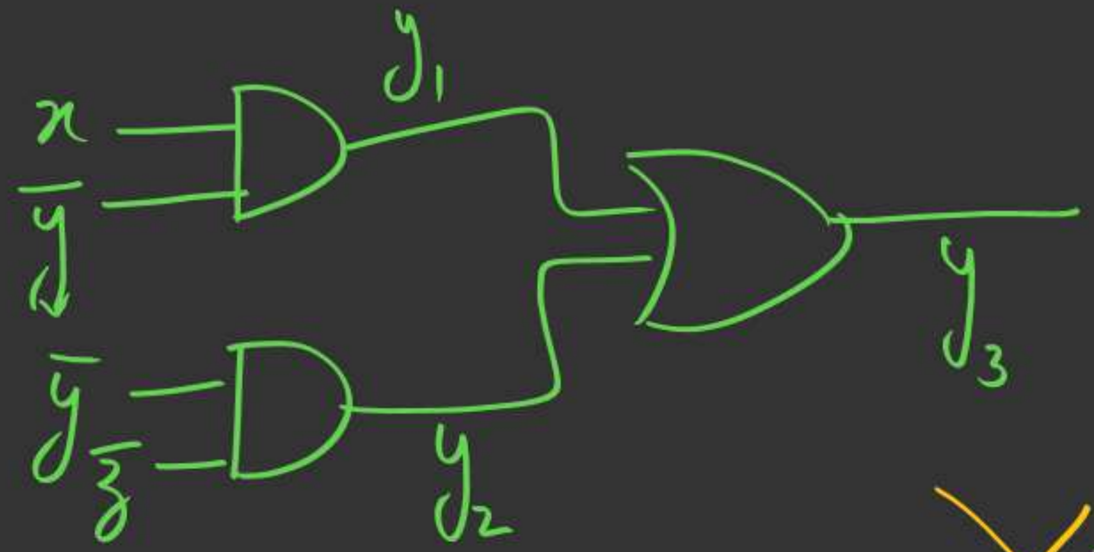
$A \oplus 1 = \bar{A}$

$A \oplus A = 0$

$A \oplus 0 = A$

$A \oplus \bar{A} = 1$

(8)



If $y = y_1 \oplus y_2 \oplus y_3$

then $y = ?$

A) $x\bar{y} + \bar{y}\bar{z}$

B) $x \oplus y \oplus z$

C) $x\bar{y}\bar{z}$

D) $x + \bar{y} + \bar{z}$

$y_1 = x \cdot \bar{y}$

$y_2 = \bar{y} \cdot \bar{z}$

$y_3 = x\bar{y} + \bar{y}\bar{z}$

$$y = y_1 \cdot y_2$$
$$= (x\bar{y}) \cdot (\bar{y}\bar{z})$$
$$= x\bar{y}\bar{z}$$

$y = y_1 \oplus y_2 \oplus y_3$

$= x\bar{y} \oplus \bar{y}\bar{z} \oplus x\bar{y} + \bar{y}\bar{z}$

lengthy

$A \oplus B \oplus (A+B) = A \cdot B$

Toppers: $y = y_1 \oplus y_2 \oplus y_3$

$y_3 = y_1 + y_2$

$y = y_1 \oplus y_2 \oplus (y_1 + y_2)$

$= y_1 \cdot y_2$

(Q) $\bar{A}(A\bar{B}+\bar{A}+C)+\bar{B}(B\bar{C}+\bar{B}C+\bar{B}\bar{C})+AB\bar{C}(B+\bar{A}B+\bar{A}C)$

$\bar{A}(\bar{A}+A\bar{B}+C)+\bar{B}(B\bar{C}+\bar{B}(C+\bar{C}))$

$P(P+\text{any})=P$

$+A\bar{C}\cdot B(B+x)$

$=\bar{A}+\bar{B}(B\bar{C}+\bar{B})+A\bar{C}B$

$=\bar{A}+\bar{B}+A\bar{C}B \quad = \bar{A}+(\bar{B}+A\bar{C})(\cancel{\bar{B}+B})$

$=\bar{A}+\bar{B}+A\bar{C} \quad = (\cancel{\bar{A}+A})(\bar{A}+\bar{C})+\bar{B}$

$=\bar{A}+\bar{B}+\bar{C} = \overline{A\cdot B\cdot C}$

(Q) $AB+BC+\bar{A}B\bar{C}D$

$B(A+C)+\bar{A}B\bar{C}D$

$=B(A+C+\bar{A}\bar{C}D)$

$=B(P+\bar{P}\cdot D)$

$=B\left(\underline{(P+\bar{P})}\cdot(P+D)\right)$ $=B\cdot(P+D)$

$=B\cdot(A+C+D)$ $=\underline{AB+B\cdot C+BD}$

Demorgan's

$A+C$ $\qquad \bar{A}\,\bar{C}$

P or $\overline{A+C}=\bar{A}\cdot\bar{C}$

(Q) $(A+\bar{B}+\bar{C}+D)(A+\bar{B}+\bar{C}+\bar{D})\cdot(\bar{A}+\bar{B}+\bar{C})$

$(P+Q)(P+R) = P+QR$

$$= \left((A+\bar{B}+\bar{C})+\cancel{D\cdot\bar{D}}\right)(\bar{A}+\bar{B}+\bar{C})$$

$$= (A+\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C})$$

$$= \left((\bar{B}+\bar{C})+\cancel{A\cdot\bar{A}}\right)$$

$$= \bar{B}+\bar{C} = \overline{B\cdot C}$$

$$(P+Q)(P+R)(P+S) = P+QRS$$

$$\bar{B}+\bar{C}+\left((A+D)(A+\bar{D})\,\bar{C}\right)$$

$$= \bar{B}+\bar{C}+\left((A+D\bar{D})\,\bar{C}\right)$$

$$= \bar{B}+\bar{C}+(A\cdot\bar{C})$$

$$= \bar{B}+\bar{C}(1+A)$$

$$= \bar{B}+\bar{C} = \overline{B\cdot C}$$

(Q) $(A+\bar{B})(\bar{B}+C)(\bar{A}+\bar{C})$

$$= (\bar{B}+A\cdot C)(\bar{A}+\bar{C}) \rightarrow \bar{B}\bar{A}+\bar{B}\bar{C}$$

$$\left\{\begin{aligned} P &= A\cdot C \\ \bar{P} &= \overline{A\cdot C} = \bar{A}+\bar{C} \end{aligned}\right\}$$

$$= (\bar{B}+P)(\bar{P}) = \bar{B}\bar{P}$$

$$= \bar{B}(\bar{A}+\bar{C}) = \bar{B}\bar{A}+\bar{B}\cdot\bar{C} = \overline{A+B}+\overline{B+C}$$

$$\bar{P} + PQR$$

$$= \underline{\underline{(\bar{P} + P)}}(\bar{P} + QR) = \bar{P} + QR = \underline{(\bar{P} + Q)(\bar{P} + R)}$$

$$\overline{P}\,\overline{Q} + \overline{Q}R + \overline{P}\,\overline{R}$$

P ×

Q ×

R

Redundant

$$= \left[\overline{P}\,\overline{R} + \overline{Q}R\right]$$

HW

validate using Truth table.

P	Q	R

$$P\bar{Q}R + QR$$

$$= R(P\bar{Q} + Q) = R((P+Q)(Q+\bar{Q}))$$

$$= R((P+Q)(1)) = \underline{PR+QR}$$

(Q) $\bar{P} \oplus \overline{(P+QR)}$

A) PQR

B) $P\,\overline{QR}$

C) $\bar{P}QR$

D) None

$\Rightarrow \bar{P} \oplus \bar{Q} = P \oplus Q$

$\bar{P} \oplus (\overline{P+QR}) = P \oplus (P+QR)$

$= \bar{P}(P+QR) + P\cdot(\overline{P+QR})$

$\cancel{\overline{P+QR} = \bar{P}\cdot\bar{Q}\cdot\bar{R}}$

$= \cancel{\bar{P}P} + \bar{P}QR + P(\bar{P}\cdot\overline{QR})$

$= \bar{P}QR + \cancel{P\bar{P}\,\overline{QR}}$

$= \underline{\underline{\bar{P}QR}}$

(Q) $\bar{P}Q + QR + \overline{PQR}$

17%

$\bar{P}Q + QR + \overline{PQR}$

$\bar{P}Q + QR + \bar{P} + \bar{Q} + \bar{R}$

$= \bar{P}(1+Q) + QR + \bar{Q} + \bar{R}$

$= \bar{P} + A + \bar{A} = \bar{P} + 1 = 1$

$A = QR$

$\bar{A} = \bar{Q} + \bar{R}$

(Q) $\bar{P}Q \oplus QR \oplus \overline{PQR}$

A) $\bar{P} + Q + \bar{R}$

B) $P + \bar{Q} + R$ ✓

C) $\bar{P} + \bar{Q} + \bar{R}$

D) $\bar{P}Q + QR$

Imp

$$\bar{P}Q \oplus QR \oplus \overline{PQR}$$

$$= \overline{\bar{P}Q \oplus QR \oplus PQR}$$

$$= \overline{Q[\bar{P} \oplus R \oplus PR]}$$

$$= \overline{Q[\overline{P \oplus R \oplus PR}]}$$

$$= \overline{Q(\overline{P+R})} = \bar{Q} + \overline{\overline{P+R}} = P + \bar{Q} + R$$

$$\overline{P \oplus Q} = \bar{P} \oplus Q = P \oplus \bar{Q}$$

$$A \oplus B \oplus AB = (A+B)$$

$$B \cdot B = B$$

$$AB \oplus BC \oplus ABC$$

$$AB \oplus BC \oplus AB \cdot BC$$

Bonus

0 1 0

$$B\,[A \oplus C \oplus AC]$$

$$= B \cdot (A + C)$$

$$= BA + BC$$

$P = AB$

$Q = BC$

$$P \oplus Q \oplus PQ = P + Q$$

$$= AB + BC$$

(Q) $AB \oplus BC = y$

AND gate $\rightarrow$ 1 unit (Rs)

XOR gate $\rightarrow$ 2 unit (Rs)

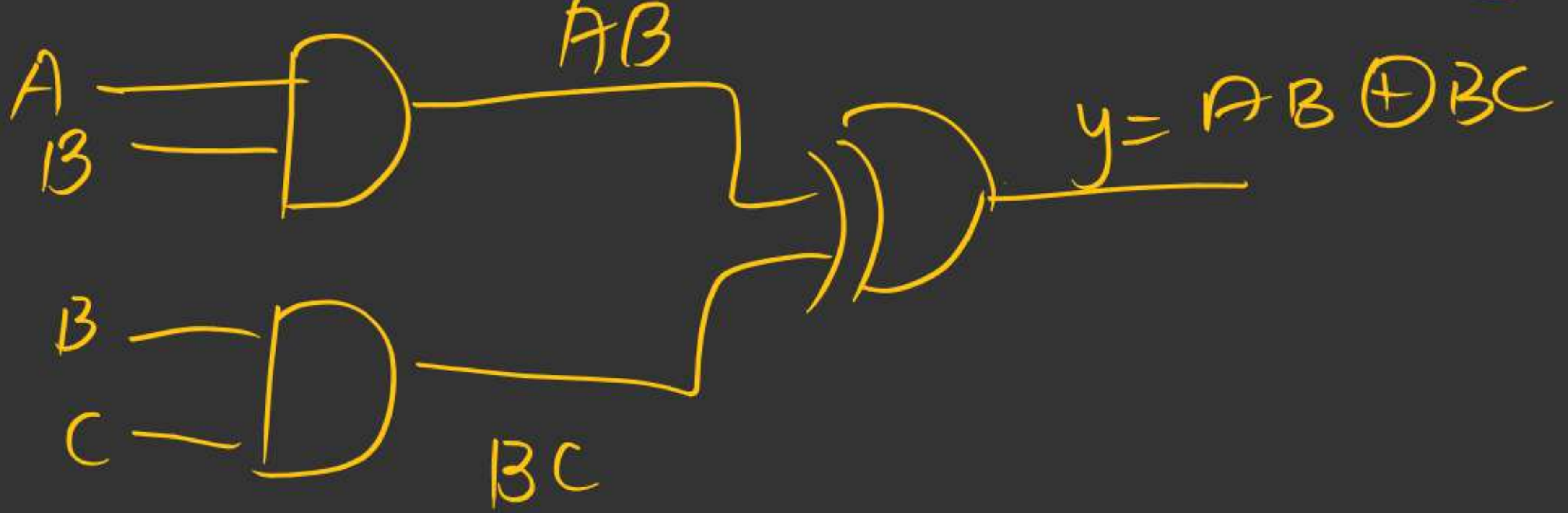
min units?

Appl

2AND + 1 XOR

$= 2 \times 1 + 1 \times 2 = 2 + 2$

$= 4$ units



$$AB \oplus BC = B \cdot (A \oplus C)$$

cost : 1 XOR + 1 AND

$= 1 \times 2 + 1 \times 1$

$= \underline{\underline{3 \text{ units}}}$

THANK - YOU